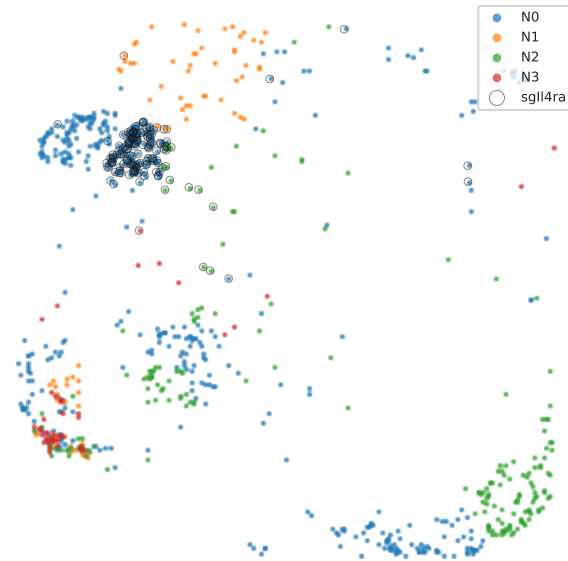


Spotlight: sgll4ra in immune microniches (extra)

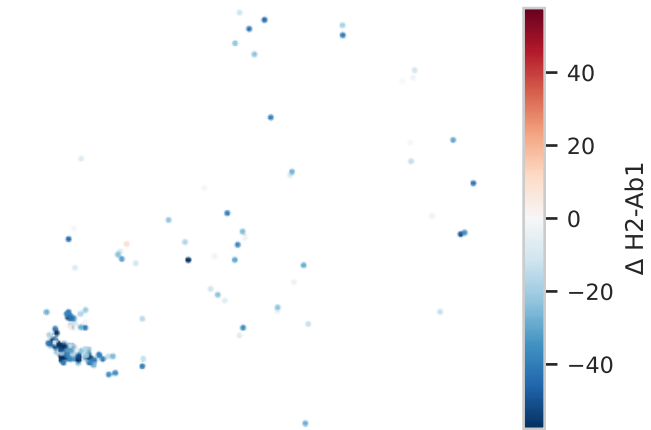
**A β -Leiden niches + sgll4ra cells
subQ-1 immune**



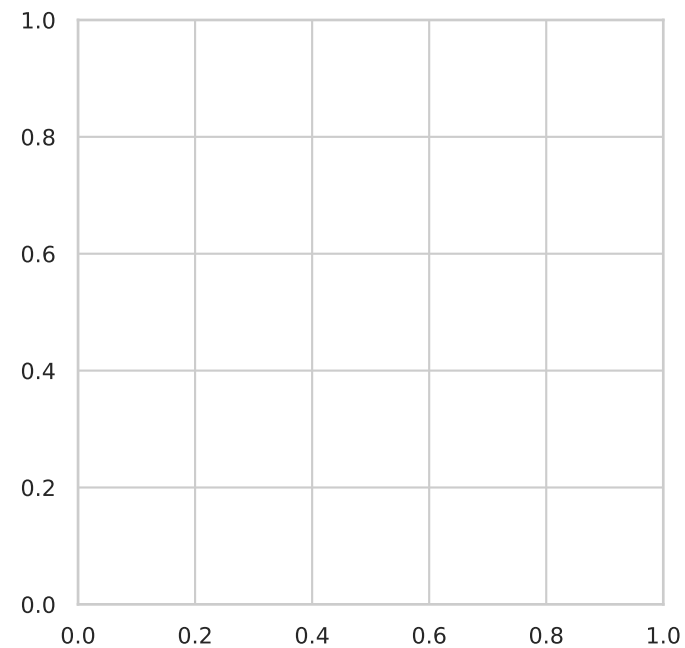
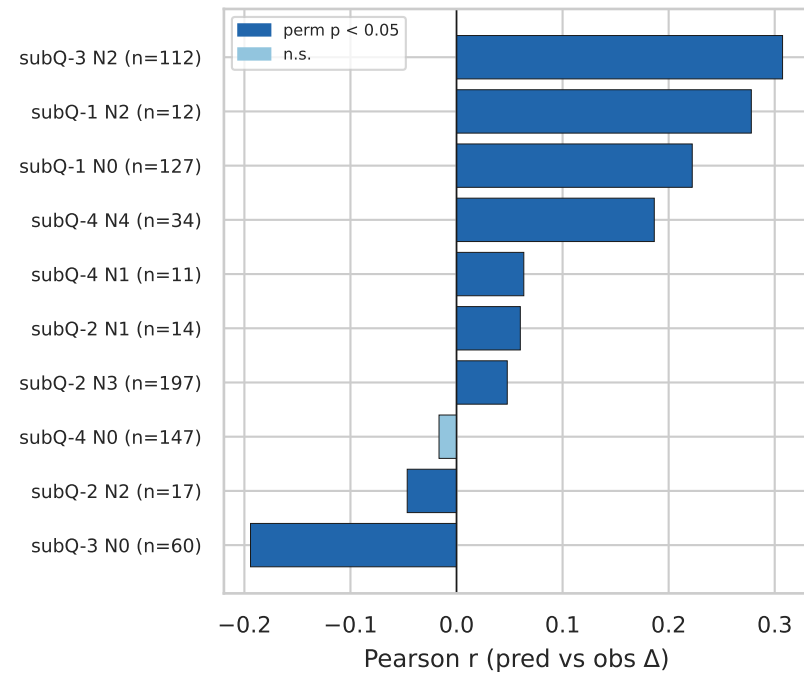
B graphclust (morphology) reference



C Predicted Δ H2-Ab1 (in-silico KO)



D Per-niche concordance (4 slices)



**F Pooled across all β -Leiden niches
 $r = +0.117$ (n=21,480 gene $\times$ niche)**

